

RESEARCH SEMINAR

SimToolReal

An Object-Centric Policy for
Zero-Shot Dexterous Tool Manipulation

Can one reusable controller
turn tool motion into robot skill?

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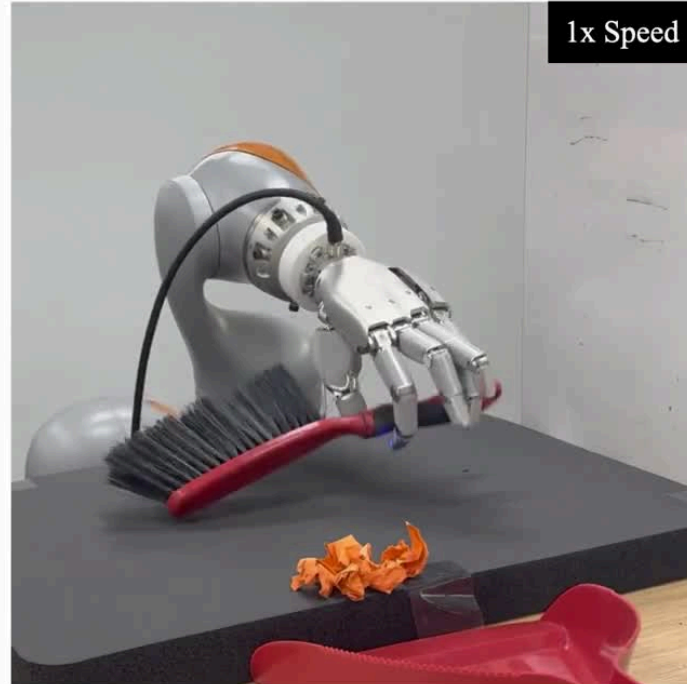
RSS 2026 · Based on arXiv v2, 24 February 2026

Real



Why is robot tool manipulation difficult?

Grasp



WHAT TO WATCH

Acquire: grasp a thin tool on a flat surface.

Reorient: move it within the fingers.

Interact: resist external forces without losing the tool.

SimToolReal project: why tool manipulation is hard · 1× speed · silent excerpt

[Original project video ↗](#)

Why not simply imitate a demonstration?

Brainstorm: what would a useful demonstration have to teach?

Teleoperate this robot

- Robot actions provide direct supervision.
- Dexterous contacts are hard to demonstrate with limited force feedback.

Retarget a human video

- Human tool use is easy to demonstrate.
- Different fingers and contacts make direct motion transfer unreliable.

Could we demonstrate the desired tool motion and let the robot learn its own contact strategy?

The transferable interface is the tool motion

TASK SPECIFICATION

Human tool motion

6D pose waypoints



Reusable goal-reaching
RL policy



ROBOT EXECUTION

Arm + hand actions

Specify where the tool goes

3D position + 3D orientation

Learn how this robot moves it

Contacts and regrasping emerge from RL

Object goals specify the motion. RL discovers robot-specific grasps and in-hand actions.

A reusable controller inside a three-stage system

1 TRAIN ONCE IN SIMULATION

Primitive tools + random goals → Physics + RL → Reusable LSTM policy

2 SPECIFY A NEW TASK

Human RGB-D demonstration

Grasp box + 6D pose waypoints

3 EXECUTE IN A CLOSED LOOP

Live tool pose + robot state

Tracking and proprioception

Perception supplies geometry

The mesh supports live pose tracking

Policy weights

Fixed LSTM actor

Current goal + grasp box

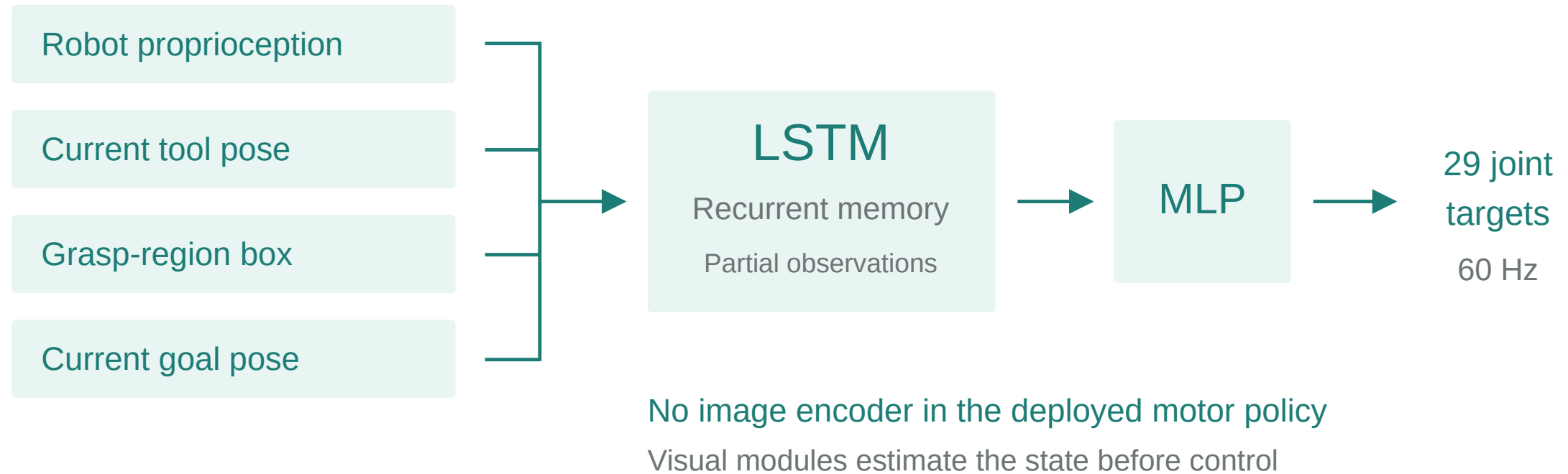
Arm + hand

29 joint targets

Physical feedback

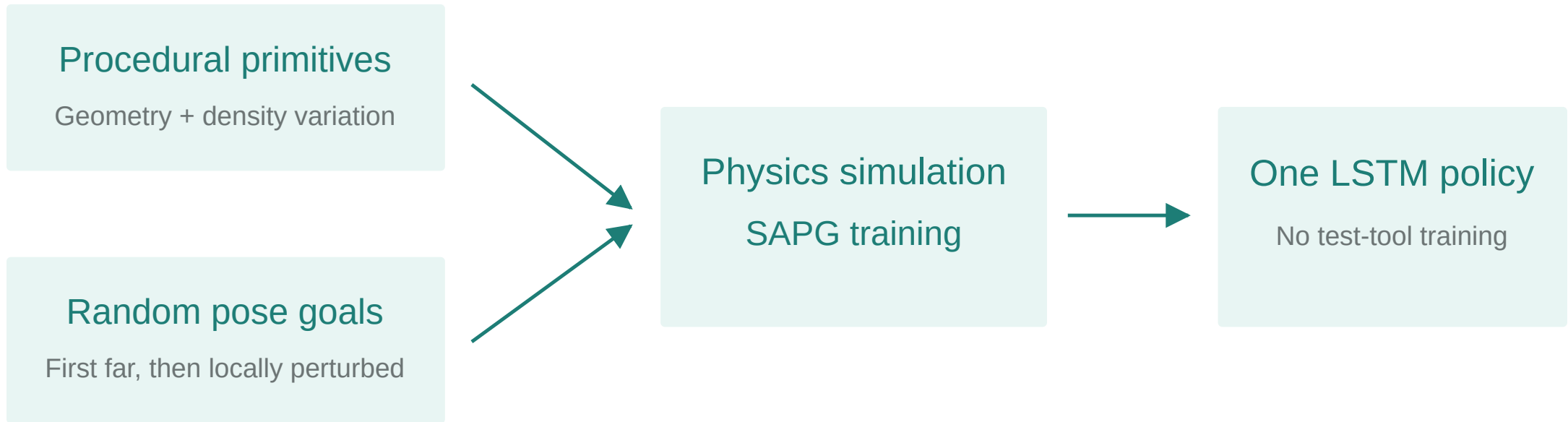
The actor converts object goals into robot actions

OBSERVATIONS



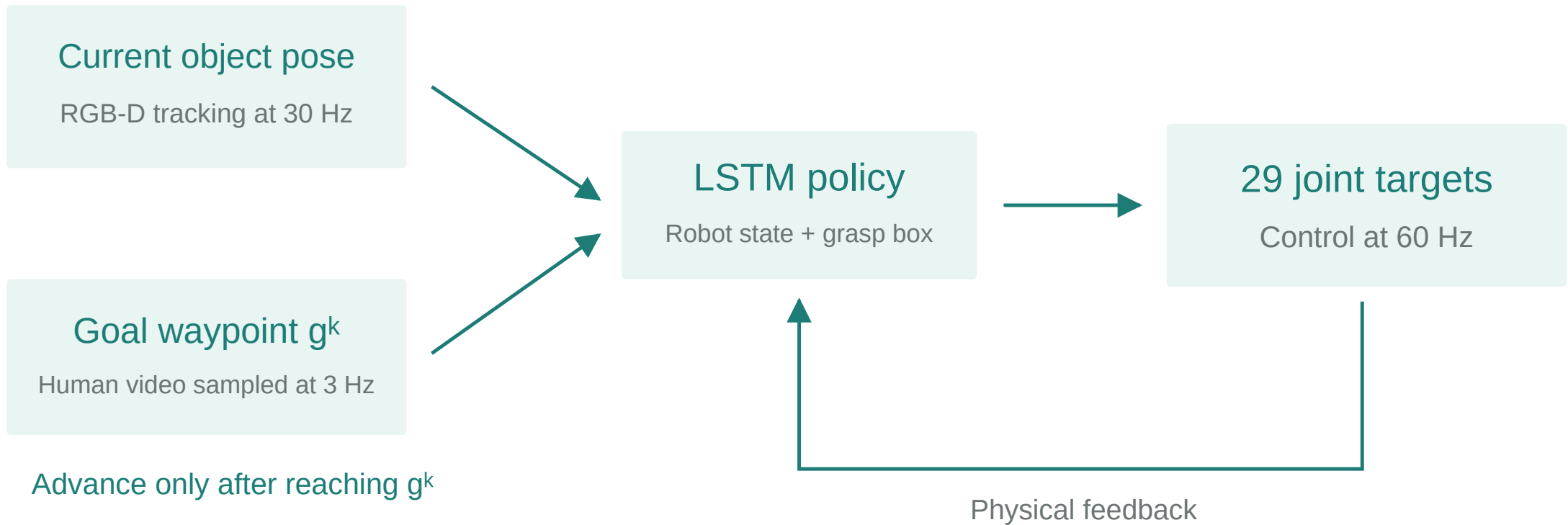
The LSTM retains interaction history while the current goal specifies what to reach.

Training varies objects and goals together



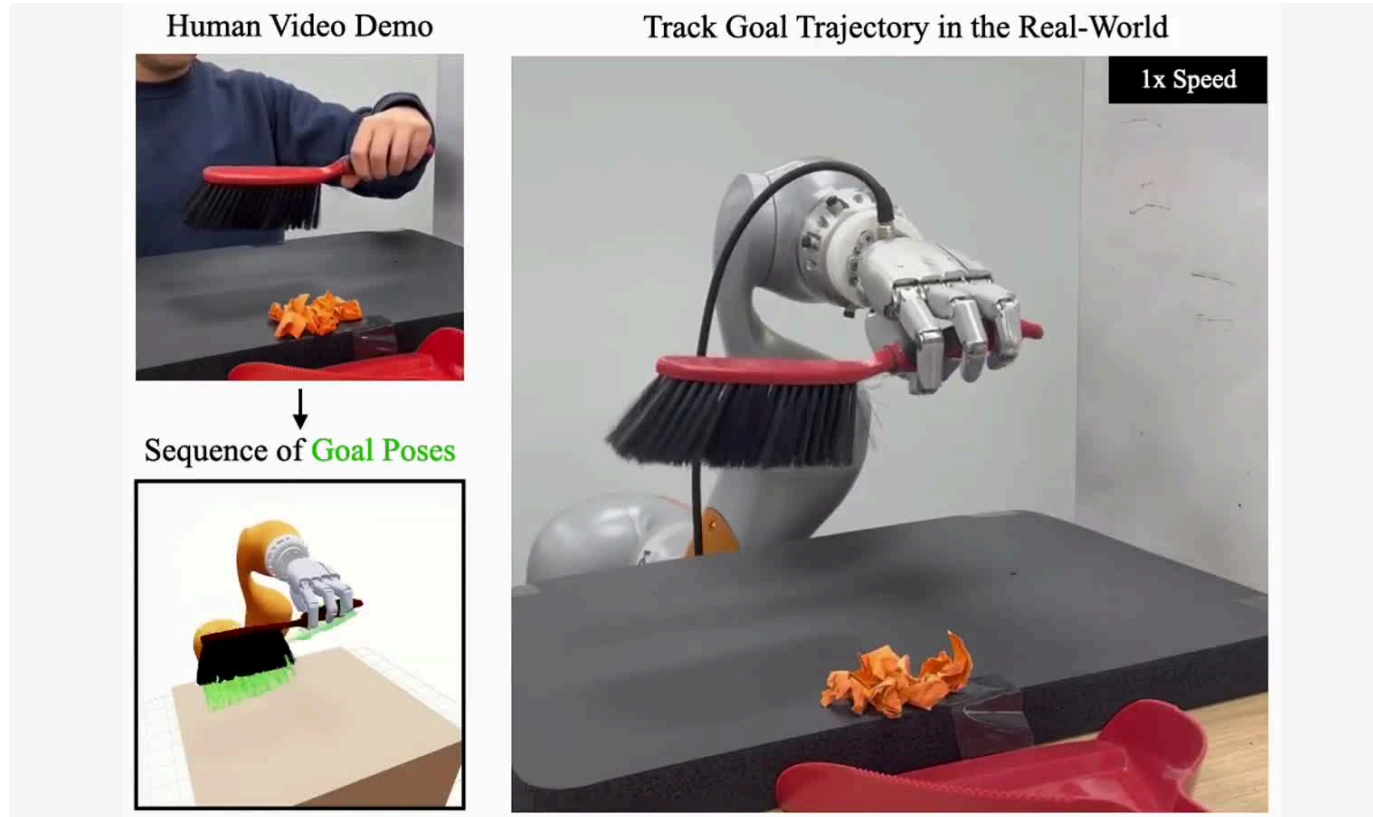
Handle–head geometry provides the prior. Grasp shaping and pose-reaching rewards teach the control.

Perception supplies state, goals advance on success



Setup: prompted SAM 2 + SAM 3D recover geometry. FoundationPose estimates the tool poses.

The closed loop follows unseen tool trajectories



SimToolReal project: real-world inference · 1x speed · silent excerpt

[Original project video ↗](#)

WHAT TO WATCH

The human demonstration supplies the **tool-motion goals**.

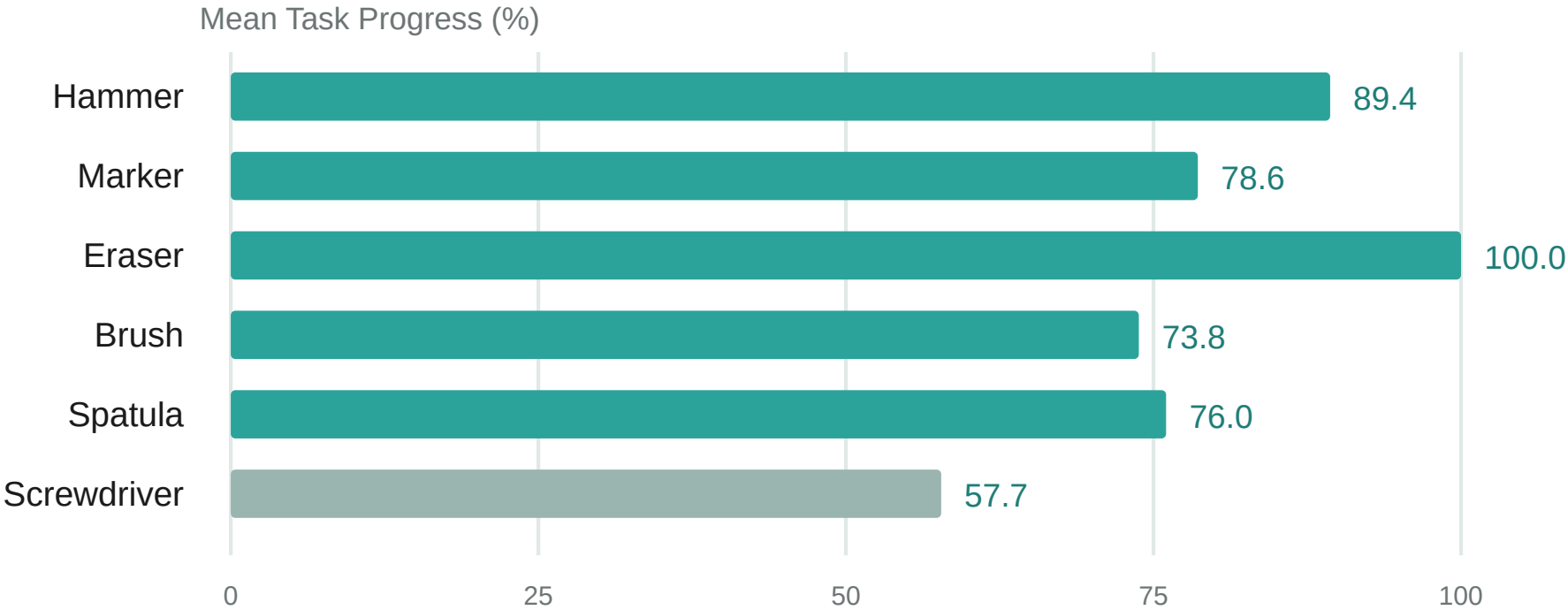
The fixed policy grasps, reorients and moves the real brush.

The goal interface specifies tool motion rather than robot finger trajectories.

Real-world transfer across six tool categories

12 objects · 24 object–task variations · 120 rollouts

Task Progress = fraction of sequential pose goals reached.



Mean progress: **79.3%**. This metric does not establish functional task success.

In-hand control matters in the brush comparison

No initial tool rotation

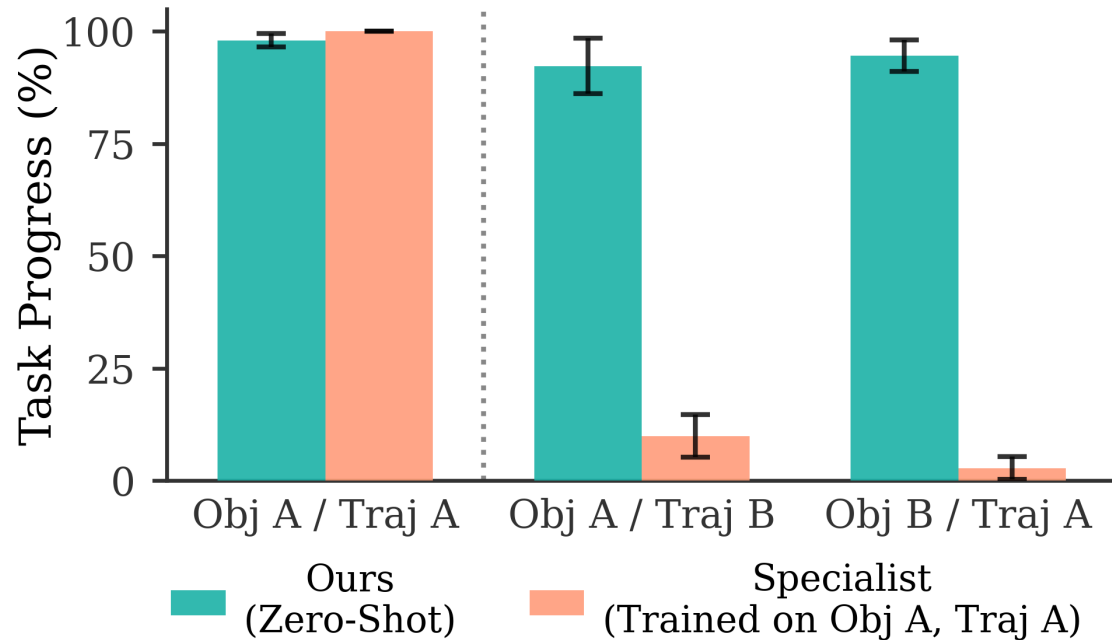


90° tool rotation required



Five rollouts per variation. Same red-brush task, two initial orientations. Values from Fig. 5.

Specialists are sensitive to changed tools and goals



Paper Fig. 6 · Comparison to specialist RL policies

- Match specialist performance on its training setup.
- Keep high progress when either the tool or the trajectory changes.
- All three conditions are evaluated in simulation.

Perception is the largest recorded failure category



■ Pose tracking loss

43.7%

43.7%

of observed failures

Perception is a system bottleneck

■ Object drops

34.5%

■ Incomplete rotation

18.2%

■ Grasp failure

3.6%

A robust controller cannot compensate for a tracker that loses the object entirely.

The pose interface leaves important state out

Assumption or omission	Where it can fail
Rigid tool state	Scissors and articulated tools need internal joint state.
Environment-blind policy inputs	Clutter can make a pose path collide with the scene.
No explicit task-force objective	Correct tool motion may not deliver the required force.
Fixed high-level goal sequence	The system does not replan the task when the world changes.

The demonstrated result is reusable trajectory execution under a structured deployment pipeline.

How far can a compact task interface take us?

PRESENTER HYPOTHESES · BEYOND THE PAPER'S EXPERIMENTS

A complementary route to general skills

Structured simulation can train reusable motor skills that a higher-level system composes.

The paper shows transfer in a structured setup. Online skill learning in the wild remains open.

Other interactions, other state

Rigid motion may use pose goals. Insertion may need force and contact. Doors may need articulation state.

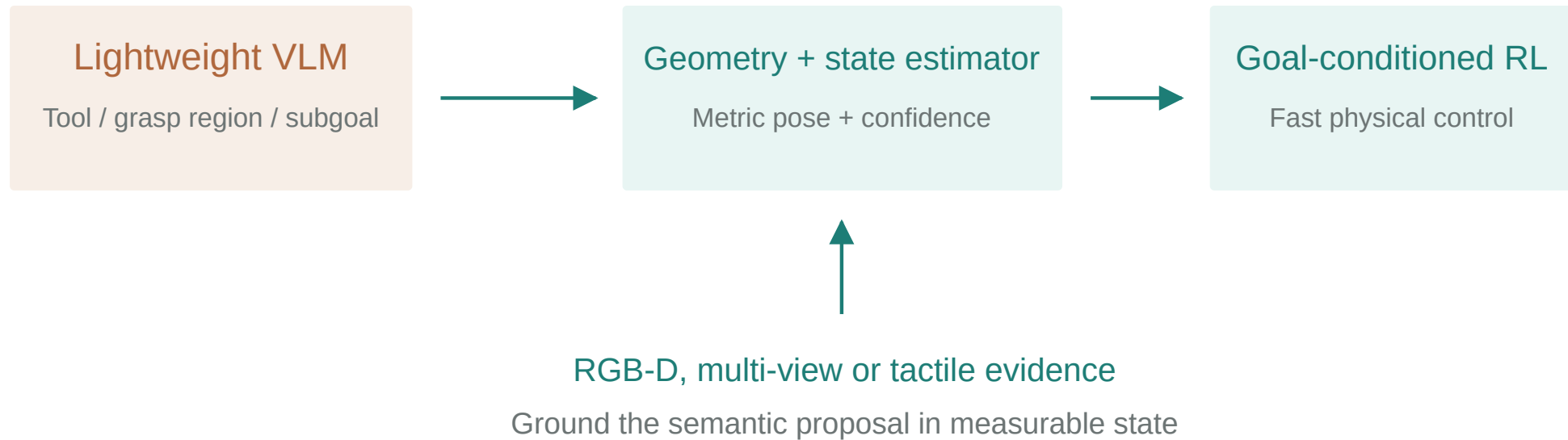
The question is which compact interface retains enough information for the task.

The generalizable idea is task-interface design. Its sufficient state depends on the interaction.

A lightweight VLM could improve the semantic interface

HYPOTHESIS 3 · PROPOSED ARCHITECTURE

PROPOSED EXTENSION



A semantic model may reduce prompting and improve goal selection. Metric, low-latency tracking still needs validation.

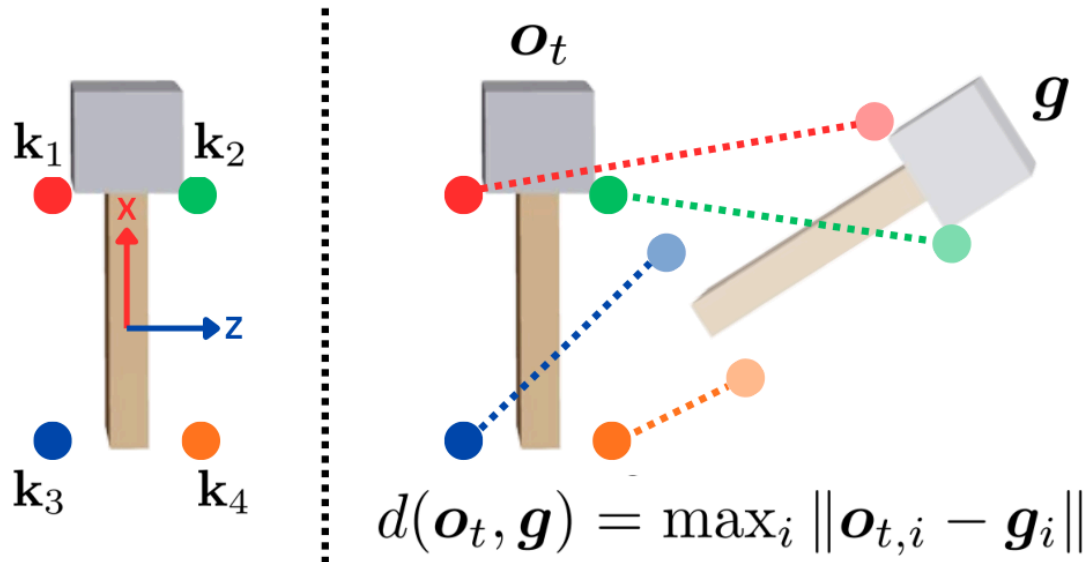
A reusable motor skill needs a well-chosen task interface

- 01 Specify motion at the object level**
Human video supplies goals without prescribing robot finger motion.
- 02 Learn the controller over varied physics**
Procedural primitives and random goals support policy reuse.
- 03 Preserve the boundary of the evidence**
Strong trajectory transfer still leaves perception, force and replanning open.

Which parts of a manipulation task can we safely compress into a reusable goal interface?

BACKUP

Pose error couples translation and rotation



Paper Fig. 9 · Four object-frame pose keypoints

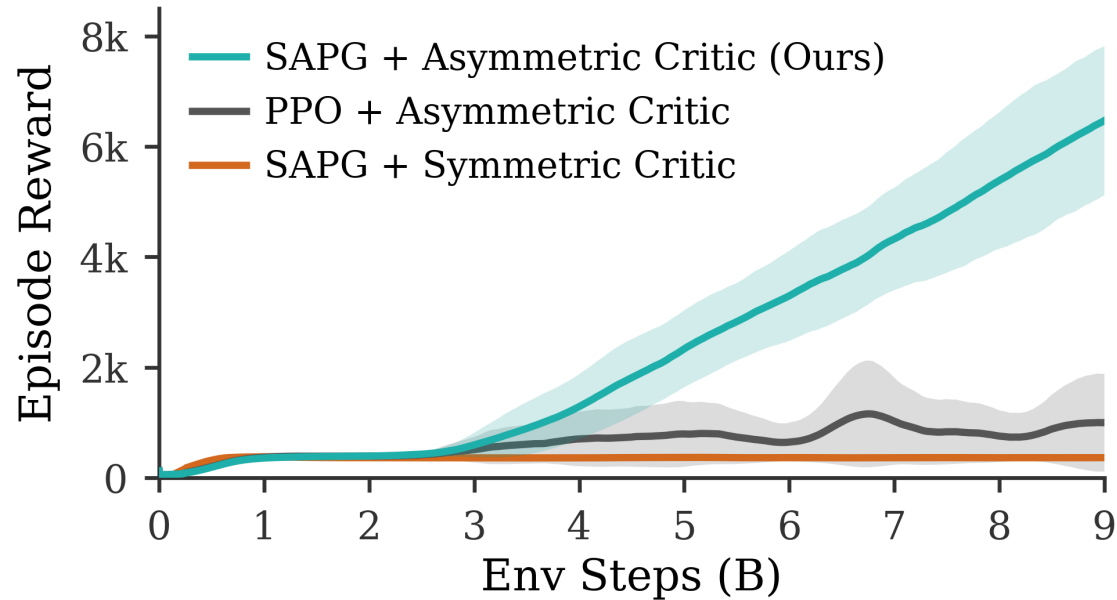
$$d(o_t, g) = \max_{i=1, \dots, 4} \|o_{t,i} - g_i\|$$

- Reward keypoints use fixed scales across objects.
- Observation keypoints depend on the object instance.
- Training tolerance: 1 cm. Evaluation: 2 cm.

A “2 cm pose error” is a keypoint-space criterion, not a separate orientation threshold.

BACKUP

Ablations identify important RL ingredients



Paper Fig. 8 · Training reward, mean over five seeds

- Replacing SAPG with PPO lowers training reward.
- Removing the asymmetric critic severely hinders learning.
- These curves cover 9B environment steps.

This ablation measures training reward, not real-world functional task completion.

Implementation details that matter for reproduction

Component	Reported configuration
Robot	7-DoF KUKA iiwa 14 + 22-DoF Sharpa left hand
Simulator / parallelism	IsaacGym / 24,576 environments
Simulation / control	120 Hz / 60 Hz
Actor	LSTM[1024] + MLP[1024, 1024, 512, 512]
Critic	MLP[1024, 1024, 512, 512], privileged simulator state
Pose observations	1 cm translation noise, 5° rotation noise, up to 10-step delay
Perturbations	5 N force scale, 0.5 Nm torque scale

References, videos and source material

Main paper

Kedia, Lum, Bohg and Liu. *SimToolReal: An Object-Centric Policy for Zero-Shot Dexterous Tool Manipulation*.
arxiv.org/abs/2602.16863v2 · [RSS 2026 proceedings](#)

Project and code

simtoolreal.github.io · github.com/tylerlum/simtoolreal

Closely related work

Chen et al. *Object-Centric Dexterous Manipulation from Human Motion Data*. CoRL 2024.

Lum et al. *Crossing the Human-Robot Embodiment Gap with Sim-to-Real RL Using One Human Demonstration*. 2025.

Petrenko et al. *DexPBT*. 2023.

Design reference

[Mizoreww/steerable-vla-talk](https://mizoreww.github.io/steerable-vla-talk) and the previous A2A seminar deck.